

SDP 302

Problem Set: Multicollinearity & PCA

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Instructions & Information

The Longley data set is a small macroeconomic data set widely used in statistics, econometrics, and data science. Introduced by Longley [1967](#) to assess computational precision, it captures the macroeconomic landscape of post-WWII US economic growth and demographic expansion across just 16 rows and 7 variables. Almost all variables exhibit near-perfect co-movement, making the data set notorious for high inter-variable correlation. Consequently, it serves both as a classic stress test for statistical modeling and a prime example for severe multicollinearity and numerical instability. Despite its small size for modern standards, the Longley data set remains popular due to its simplicity and accessibility. It comes pre-packaged in many statistical libraries and features in intermediate and advanced textbooks, including ,D. Gujarati, D. N. Gujarati, and Porter [2009](#) Greene [2012](#) or Heiberger and Holland [2015](#), amongst others.

Based on macroeconomic demand theory, Longley used the following multiple linear regression equation to model post-WWII US employment:

$$\begin{aligned} \text{TOTEMP}_t = & \beta_0 + \beta_1 \text{GNPDEFL}_t + \beta_2 \text{GNP}_t + \beta_3 \text{UNEMP}_t + \beta_4 \text{ARMED}_t \\ & + \beta_5 \text{POP}_t + \beta_6 \text{YEAR}_t + \epsilon_t \end{aligned}$$

When standard OLS regression is applied directly to this data set, strong correlations among the independent variables lead to inflated variance, unstable coefficient estimates, and numerical instability. *Principal Component Analysis (PCA)* is an unsupervised technique used to address these issues. It simplifies high-dimensional data by transforming correlated variables into a smaller set of uncorrelated variables called the principal components, which retain as much of the original variance and information as possible. Constructed as linear combinations of mean-centred variables using eigenvectors, principal components are orthogonal, i.e. statistically independent, by design. As a result, PCA also helps to uncover underlying structural relationships that might otherwise remain hidden. For more information on PCA, the book by James et al. [2013](#) provides a thorough treatment of both the theoretical framework and practical use cases. In Python, PCA is part of the `statsmodel` library.

References

- Greene, William H. (2012). *Econometric Analysis*. 7th. Boston: Prentice Hall.
- Gujarati, Damodar, Damodar N. Gujarati, and Dawn C. Porter (2009). *Basic Econometrics*. 5th. McGraw-Hill.
- Heiberger, Richard M. and Burt Holland (2015). *Statistical Analysis and Data Display*. 2nd. Springer.
- James, Gareth et al. (2013). *An Introduction to Statistical Learning*. Vol. 103. Springer Texts in Statistics. New York, NY: Springer New York.
- Longley, James W. (1967). “An Appraisal of Least Squares Programs for the Electronic Computer from the Point of View of the User”. In: *Journal of the American Statistical Association* 62.310, pp. 319–341.

[EXR] Exercise 1 (OLS & Multicollinearity: The Longley Data Set)

The Longley data set is part of Python's `statsmodel` library and can easily be loaded calling the method `statsmodel.datasets.longley.load_pandas()`. Import the data into a `DataFrame` and answer the following questions:

- a) Display the data in a well-formatted table. What structural patterns or trends do you observe across the variables?
- b) Based on macroeconomic theory, what are the expected signs (+ or -) for each coefficient in Longley's regression equation?
- c) Fit the OLS regression model. Do the estimated coefficients match your expected signs? Do you notice any surprising results such as unexpected p-values or signs? What might explain them?
- d) Compute and display the correlation matrix for the six independent variables. Also for each variable calculate the variance inflation factor. What conclusions can you draw about variable redundancy?
- e) Normalise each independent variable using 1947 as the base year. Generate matrix scatter plots of all pairwise combinations, and plot the trends of all six variables from 1947 to 1962 in a single chart. What patterns emerge?
- f) Remove the two variables `UNEMP` and `YEAR` and repeat the OLS regression from above. Is this step backed by statistical or theoretical design justifications?
- g) Apply PCA to reduce dimensionality and eliminate correlation. Can you offer a meaningful interpretation for each of the primary principal components?